Rea-Sixty User Manual

Native REAPER ↔ SSL UF8 / UC1 driver

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Contents

1 Introduction

1.1 What Rea-Sixty is

Rea-Sixty is a REAPER extension that drives the SSL UF8 and UC1 control surfaces directly from REAPER. It replaces SSL 360° on the host side. One extension file installs into REAPER's `UserPlugins` directory; SSL 360° no longer needs to run, the surface is no longer behind a virtual MIDI port, and the per-track SSL plug-in that SSL 360° requires for track colours is no longer needed.

What goes out on the USB wire is the same byte protocol SSL 360° uses, re-emitted by REAPER. No SSL binaries, firmware, or trademarks are redistributed.

1.2 What you need

- REAPER on macOS (Apple Silicon), Windows (x64), or Linux (x86_64). Tested against REAPER 6 and 7 through 7.71.
- An SSL UF8 plugged in over USB-C. UC1 is supported optionally; UF8-only or UC1-only rigs are fine.
- **ReaImGui** (install via ReaPack from Extensions → ReaPack → Browse packages → ReaImGui). Without ReaImGui the Settings window stays empty, but hardware control still works.
- **SSL 360° must not be running.** It claims the UF8/UC1 vendor interface exclusively. If it is running when REAPER starts, the surface will not appear and REAPER's Console shows an error.

Runtime dependencies (`libusb`, `hidapi`) ship inside the platform archives; no separate install needed.

1.3 Versioning

This manual documents Rea-Sixty v0.1.8. Earlier manuals (anything dated before 2026-05-26) are superseded.

2 Installation

2.1 Via ReaPack (recommended)

In REAPER:

1. **Extensions → ReaPack → Manage repositories → Import/export → Import repositories**
2. Paste: `https://github.com/acklin83/reaper-scripts/raw/main/index.xml`
3. **Browse packages** → filter `Rea-Sixty` → **Install**
4. Restart REAPER
5. **Preferences → Control/OSC/Web → Add → Rea-Sixty**

First-run setup buttons live in **Settings → About**:

- **Windows:** *Install UF8/UC1 WinUSB driver* (one UAC prompt)
- **Linux:** *Install Linux udev rule* (one pkexec prompt)
- **macOS:** no setup needed; IOKit already grants libusb access to the device class

2.2 Manual install

Download from https://github.com/acklin83/Rea-Sixty/releases:

- **Mac:** `rea-sixty-mac-v0.1.7.zip` — three Apple-notarised dylibs. Unzip into `~/Library/Application Support/REAPER/UserPlugins/`.
- **Windows:** `rea-sixty-win-v0.1.7.zip` — three DLLs. Unzip into `%APPDATA%\REAPER\UserPlugins\`.
- **Linux:** `rea-sixty-linux-v0.1.7.tar.gz` — `.so` + udev rule + INSTALL.txt. Follow INSTALL.txt.

2.3 Enabling the surface

After install, restart REAPER, then:

Preferences → Control/OSC/Web → Add → Rea-Sixty

No MIDI device assignment needed — the extension claims the UF8/UC1 over USB on its own.

2.4 Uninstall

- **Via ReaPack:** Browse packages → right-click Rea-Sixty → Uninstall.
- **Manual:** delete `reaper_rea-sixty.{dylib,dll,so}` plus the two runtime libraries (`libusb`, `libhidapi/hidapi`) from REAPER's UserPlugins directory. Remove the Control Surface entry under Preferences → Control/OSC/Web.

3 Concepts

A handful of terms appear throughout the manual.

3.1 Focused track

The track REAPER considers "selected first" — `GetSelectedTrack(nullptr, 0)`. UC1 follows it by default; the SSL Channel Strip / Bus Compressor section reads the focused track's plug-in.

3.2 FX vs Instance

- **FX** = any audio effect on a REAPER track (everything `TrackFX_GetCount` sees).
- **Instance** = the surface-mapped subset only:
  - SSL Channel Strip 2 + 4K B/E/G variants
  - SSL Bus Compressor 2
  - SSL 360° Link
  - Combo plug-ins of the above
  - User-mapped UF8-only plug-ins with `uf8Mode` set in their FX Learn catalog entry

Every Instance is an FX. Most FX are not Instances.

The distinction matters for two pairs of cycle actions:

- **FX Cycle** walks every FX on the track.
- **Instance Cycle** walks only Instances on the track.

Six action names follow the same rule (per-surface × per-scope × per-cycle-kind). See the Cycle actions section under Native actions.

3.3 FX-Cursor

A per-track pointer at the last FX a cycle landed on (the index the next cycle step proceeds from). Persistent across mode changes. When a cycle lands on a learned Instance, the cursor also updates the per-domain Instance index so SSL Strip Mode / UF8 Plug-in Mode follow.

3.4 Plug-in identity

When you rename an FX instance in REAPER's FX chain ("Townhouse Comp" instead of the factory name), Rea-Sixty uses the rename everywhere the colour-bar would otherwise show the factory short name. Internally the rename is resolved by FX-GUID, so reordering or chunk-replacing the FX preserves the binding to the identity.

3.5 Domain

The Instance domain (`ChannelStrip`, `BusComp`, or `None` for UF8-only user maps). Drives which UC1 section refreshes when the Instance cursor moves.

4 Selection Modes

Each strip's `SEL` button is hijacked by the active Selection Mode. Toggle modes via the dedicated buttons on the UF8 (mapped through Bindings) or via the per-mode builtin actions.

4.1 NORM

The default — `SEL` toggles REAPER track selection. V-Pots default to Pan. No automatic colour-bar overrides on the upper LCD.

4.2 REC

Each strip's `SEL` toggles arm. V-Pots stay on Pan unless `REC + RME` integration is on (next mode).

4.3 REC + MON

Each strip's `SEL` toggles arm + monitor. Otherwise like REC.

4.3.1 REC + RME (TotalReaper integration)

A sub-mode of REC / REC+MON. When **Settings → Modes → REC → "Enable RME / TotalReaper integration"** is on and TotalReaper is installed:

- V-Pot rotation = preamp gain ±1 dB (per the *V-Pot rotation → Preamp gain* toggle)
- V-Pot push / Cut / Solo / Polarity = configurable TotalReaper actions (48V toggle, pad toggle, phase invert, AutoLevel toggle).
- Shift+V-Pot rotation = change input channel
- The strip's colour bar shows the input name ("Mic 1", "Line 3") instead of the CS variant label

4.4 AUTO

The strip's `SEL` cycles through automation modes (Off → Read → Touch → Latch → Write → Trim, then back). V-Pots become read-only automation-mode indicators.

When **Settings → Modes → AUTO → "Hide Trim/Read tracks while in AUTO mode"** is on, tracks set to Trim or Read disappear from the surface so only writing-armed tracks remain visible.

Leaving AUTO restores the previously-visible track list and refocuses the surface on the selected track.

If a Selection Set is active and **Settings → Modes → AUTO → "Selection-Set auto-mode"** is set to a value other than `None`, recalling a selset in AUTO mode auto-arms the set's tracks to that automation mode. Leaving AUTO reverts those tracks to Trim/Read.

4.5 FX Cycle (V-Pot Sel-Mode)

Each strip's V-Pot rotation walks every FX on the strip's track. Push opens the active FX's floating window. The strip's colour bar shows the FX-cursor's current FX name. Internal name `SelectionMode::Instance` — kept for binding-file compatibility despite walking all FX.

4.6 Instance Cycle (V-Pot Sel-Mode)

Each strip's V-Pot rotation walks only Instances (CS / BC / UF8-Mode-learned) on the strip's track. Push opens the active Instance's floating GUI. No-op when the strip's track has fewer than 1 Instance.

When the cycle lands on a learned Instance the per-domain Instance index updates so SSL Strip Mode + UF8 Plug-in Mode + the UC1 sections follow.

5 Channel Encoder modes

The large notched CHANNEL encoder (right of the strips, pushable, surrounded by the cursor pad) runs one of nine modes. Switch with the corresponding `encoder_*` builtin. The current mode persists across REAPER restarts.

Mode	Rotation acts on	Notes
Channel Select	Move REAPER track selection ±	Default mode. Strips re-bank to keep selection visible.
Nudge	Playhead nudge	Step size from REAPER's nudge setting
Mousewheel	Synthesised scroll-wheel under the mouse cursor	Use to scroll plug-in windows or Project Browser
Markers	Step prev / next marker	Stops playback while seeking
Bank by 1ch	Shift the surface 1 strip left/right	Sub-bank precision
Last Touched Param	Step the last-touched REAPER param ±	Fine increments
Instance (Instance Cycle)	Walk Instances on the focused track	Same dispatch as the V-Pot Sel-Mode Instance Cycle, focused-track scope
FX Cycle	Walk every FX on the focused track	Focused-track scope
Selset Cycle	Step through populated Selection Set slots (off → 1 → 2 … → off)	Skips empty slots

`Shift` + rotation re-banks ±1 strip in every mode (alias for Bank by 1ch).

The encoder also drives **SEL-Mode cycle** when **Settings → Modes → FX / Cycle → "UF8 Channel Encoder"** is ticked AND a cycle-kind Selection Mode is engaged. In that override the encoder dispatches via `reasixty_dispatchSelModeCycle` instead of its normal mode.

6 Nav Mode (Markers + Regions)

Nav Mode is a surface overlay — separate from the Selection Modes — that turns the UF8 strips into a live marker / region jump panel. It can be active alongside any Selection Mode; toggling it does not change `g_selectionMode`.

6.1 Engaging Nav Mode

Three bindable builtins:

- `marker_overlay_toggle` — toggle on / off, opens the default view (set in Settings → Modes → NAV).
- `marker_overlay_markers_only_toggle` — toggle on / off, locks the view to **MarkersAll** (no drill into regions).
- `marker_overlay_regions_only_toggle` — toggle on / off, locks the view to **Regions** (region presses jump only, no drill).

Bind any of these to a UF8 / UC1 button in Settings → Bindings. The same button toggles Nav Mode off again.

6.2 Three views

- **Regions** — each strip is one REAPER region. Top-soft-key press jumps the transport to that region's start (and, by default, drills into its markers).
- **MarkersInRegion** — markers inside the region the playhead is in. Auto-rolls into the next region when the playhead crosses out.
- **MarkersAll** — flat list of every marker in the project.

The default view on Nav-Mode entry is configurable (Settings → Modes → NAV → *Default view*): Regions / MarkersInRegion / MarkersAll / Last used.

6.3 What the UF8 strips show

Per strip while Nav Mode is active:

- **Top-soft-key** — press = jump (and, in Regions view, optionally drill). LED colour = the region / marker's REAPER colour (or grey if *Color-bar source: Force palette grey* is set).
- **Scribble strip upper row** — the marker / region name.
- **Scribble strip lower row** — configurable: Off (V-Pot value preserved) / Index (`R03`, `M07`) / Timecode (`MM:SS`).
- **Colour bar** — the marker / region's colour (or palette grey, per setting).

6.4 Paging

When the project has more items than 8 strips, the overlay pages 8 at a time. Pagination via:

- **CHANNEL encoder rotation** while Nav Mode is active — pages forward / backward.

6.5 UC1 Encoder 2 takeover

Off by default. When Settings → Modes → NAV → *Take over UC1 Encoder 2* is on:

- **Rotation** moves the cursor through items (sets a cursor pin that suppresses auto-follow until the playhead catches up or you push).
- **Plain push** — Jump + Drill / Jump only / Drill only (configurable).
- **Shift + push** — Drill / Back / Toggle View (configurable).
- **Long-press** (~500 ms) — Back / Add marker at playhead / Disabled (configurable).
- The UC1 central LCD switches to a marker carousel showing prev / curr / next items.

While a view-lock toggle (Markers-only / Regions-only) is engaged, shift and long-press are suppressed — only plain push fires.

6.6 Auto-Follow

Settings → Modes → NAV → *Auto-Follow playhead / edit cursor* (checkbox).

When on, the cursor strip tracks whichever marker / region the playhead is on. In MarkersInRegion view, the overlay auto-rolls into the next region when the playhead crosses out (only after the playhead was first observed inside the current filter region — suppresses the snap-back when you drill manually during playback).

Manual cursor movement (UC1 Encoder 2 rotation) pins the cursor and pauses auto-follow until the playhead catches up, or you commit (push), exit Nav Mode, drill, or change view.

6.7 Region-press behaviour

Settings → Modes → NAV → *Region-press behaviour*:

- **Jump + Drill** (default) — press a region's top-soft-key → transport jumps to region start AND the overlay drills into that region's markers.
- **Jump only** — transport jumps; overlay stays on the Regions view.
- **Drill only** — overlay drills; transport stays put.

RegionsOnly view-lock always suppresses Drill regardless of this setting.

7 UF8 hardware

The UF8 has **no transport keys, no Layer LED on Layer 3 on some units, no jog wheel**. The layout below mirrors SSL's published reference (User Guide Rev 11, p.14-17).

7.1 Strips (×8)

Per strip, from top to bottom:

Control	Function (NORM Selection Mode)	Notes
Top soft-key	SSL Soft-Key for this strip in the current PAGE bank	Default builtin `ssl_softkey(strip)`. Rebindable.
Colour TFT (scribble strip)	Upper zone = track name / mode-dependent. Lower zone = parameter readout. Track-colour bar at the bottom.	Hijacked by Plug-in Modes for parameter / FX names.
V-Pot rotation	Pan	Re-maps per Selection Mode (REC + RME → preamp gain; AUTO → automation indicator; FX Cycle / Instance Cycle → walk FX/Instances).
V-Pot push	Centre Pan	In FX Cycle / Instance Cycle Sel-Modes → open the active FX's GUI.
`SOLO`	Solo	LED colour follows REAPER track colour when *Settings → Device → Display behaviour → "SEL LED follows REAPER track colour"* is on.
`CUT`	Mute	
`SEL`	Selection-Mode dependent	NORM = exclusive select; REC = arm; REC+MON = arm + monitor; AUTO = cycle automation mode. Long-press on a folder parent toggles spill.
Capacitive touch (fader)	Drives REAPER's "touch" automation	Alt/Option held during touch → snap back on release (Settings → Device → Keyboard Options).
100 mm motorised fader	Track volume	Full 16-bit pitch-bend protocol.

7.2 Above the strips: 8 Top Soft-Keys + bank selectors

A row of 8 buttons above the V-Pots = **Top Soft-Keys 1..8**. By default each one focuses the SSL plug-in parameter at its strip position within the current PAGE bank (SSL 360° factory behaviour). Rebindable per-button.

Left of the soft-keys: 6 small bank-selector buttons — **V-POT** + **1 / 2 / 3 / 4 / 5**. Each picks one of the 6 SSL CS soft-key banks (or 2 BC banks while in Bus-Comp). Default builtin `softkey_bank_select(N)`.

7.3 Right of the strips: CHANNEL encoder + cursor pad

A single block:

- **Large notched CHANNEL encoder** (push-button rotary). Rotation drives the active Channel Encoder mode (see chapter Channel Encoder modes). Push = mode-specific.
- **Cursor pad** — 5 buttons (4 arrows + central circle) **surrounding the encoder**.
  - Default behaviour: **zoom** via the `zoom_up` / `zoom_down` / `zoom_left` / `zoom_right` / `zoom_center` builtins (REAPER actions 40112 / 40111 / 1011 / 1012 / 40295).
  - SSL's reference UG also documents a "Cursor-Transport" mode (press-and-hold CHANNEL encoder to enter; ↓=Stop ↑=Play ←=Rew →=FF centre=Rec). Rea-Sixty leaves these as the standard zoom bindings — rebind them to transport actions via Settings → Bindings if you want SSL's behaviour.
- **NAV / NUDGE / FOCUS** mode buttons (around the encoder; ButtonId entries `Nav`, `Nudge`, `EncFocus`). Default builtins switch the encoder to that mode.

Above the encoder block: **Q1 / Q2 / Q3** ("Quick" user keys). Default bindings — Q1 = CS, Q2 = BC, Q3 = I/O meter (matches SSL's locked Plug-in Mixer assignment).

7.4 Right of the encoder column: NORM / REC / AUTO

Three buttons (Selection Mode block). Default unbound — assign the `selection_mode_norm` / `selection_mode_rec` / `selection_mode_auto` builtins (or any other Selection Mode builtin, including `selection_mode_instance` / `selection_mode_instance_cycle`) via Settings → Bindings.

7.5 Above NORM/REC/AUTO: AUTOMATION row

Six buttons — **Read / Write / Touch / Latch / Trim / Off**. Default builtins `auto_read`, `auto_write`, `auto_touch`, `auto_latch`, `auto_trim`, `auto_off` (set the automation mode of the focused track).

7.6 Below the strips: PLUGIN / CHANNEL / mode row

Button	Default
`PLUGIN`	Toggles SSL Strip Mode (`ssl_strip_mode_toggle`). With Shift held: `ssl_strip_mode_toggle_with_gui`.
`CHANNEL`	`home` — clear send / receive routing toggles so V-Pots / faders return to track volume + pan.
`BANK ←` / `BANK →`	Scroll ±8 strips. In UF8 Plug-in Mode → flip between fader-banks A / B (for 16-strip plug-ins).
`PAGE ←` / `PAGE →`	Step the SSL Soft-Key PAGE bank prev / next (6 CS banks + 2 BC banks).
`SEND / PLUGIN 1..8`	8 buttons. Default `send_this(N)` / `recv_this(N)` — toggle the matching send / receive view.

7.7 FLIP / PAN / FINE

Three buttons in their own cluster:

Key	Default action
`FLIP`	`flip` — swap fader and V-Pot values for the active mode.
`PAN`	`pan_force` — force V-Pots to Pan regardless of the active Selection Mode. Escape hatch from cycle / REC / AUTO modes.
`FINE`	`mod_shift` modifier (the SSL "FINE" = "Shift" key). Double-click latches it on; press again to unlatch. With *Settings → Device → Keyboard Options → Shift activates Fine mode* on, holding this also drops V-Pot / encoder step size to ×0.25 (faders unaffected).

7.8 Layer keys

`LAYER 1 / 2 / 3` — three SSL DAW layers. Bindable. Layer 3 LED on certain UF8 units does not light up — confirmed hardware quirk, not a Rea-Sixty bug. Layer functionality itself works.

7.9 360° key

Default binding `mixer_toggle` — opens / closes the Rea-Sixty Settings window.

7.10 Foot-switch jacks

`FS1` / `FS2` on the back (1/4” TS, normally-closed momentary). No factory bindings — assign via Settings → Bindings (ButtonId `Foot1` / `Foot2`).

7.11 What the UF8 does NOT have

For clarity (Rev 11 reference, p.14-17):

- No dedicated `Play / Stop / Record / Loop` keys. Transport-on-cursor-pad is a press-and-hold-CHANNEL feature; Rea-Sixty leaves the cursor pad on zoom by default.
- No jog wheel. The CHANNEL encoder is the only large rotary.
- No master fader.
- No HUI/MCU display lane apart from the 8 strip scribble LCDs.

8 UC1 hardware

The UC1 mirrors the SSL Channel Strip 2 + Bus Compressor 2 controls on a dedicated unit, plus a central control panel for navigation. When UC1 is plugged in alongside UF8, it auto-engages on the focused REAPER track and follows the focused Instance.

The UC1 has no hardware mode-switch — the Channel Strip and Bus Compressor sections are always live, each driving whichever CS or BC Instance is currently in focus on the focused track.

8.1 Channel Strip section — left side (EQ + Filters, 12 knobs)

Top to bottom on the left half:

- **LPF / HPF** — two filter knobs (low-pass + high-pass frequency).
- **HF** — Gain + Freq (2 knobs).
- **HMF** — Gain + Freq + Q (3 knobs).
- **LMF** — Gain + Freq + Q (3 knobs).
- **LF** — Gain + Freq (2 knobs).

Plus the EQ-section buttons: **HF Bell** (HF shape), **EQ Type** (E vs G EQ curve), **EQ In** (EQ section bypass), **LF Bell** (LF shape).

8.2 Channel Strip section — right side (Dynamics + Channel, 7 knobs + 7 buttons)

- **Compressor:** Threshold + Ratio + Release (3 knobs).
- **Gate / Expander:** Threshold + Range + Hold + Release (4 knobs).

Buttons:

- **Fast Att Comp** — fast attack on the compressor.
- **Peak** — peak detection.
- **Dyn In** — Dynamics section bypass.
- **Expand** — switch the gate into expander mode.
- **Fast Att Gate** — fast attack on the gate.

Channel section (lowest row of buttons):

- **Polarity** (phase invert)
- **SC Listen** (side-chain monitor)
- **Solo Clear**
- **Solo** / **Cut** (track operations — routed to REAPER's track ops)
- **Channel In** (channel input enable)
- **Fine** (FINE / Shift modifier on UC1)

8.3 Channel Strip top row (2 knobs)

- **Input Trim** (CS input gain)

- **Channel Fader Level** (CS fader stage)

When SSL Strip Mode is engaged on the UF8, the **Channel Fader Level** parameter is what the UF8 motorised faders drive.

8.4 Bus Compressor section (7 knobs + 1 button)

- **Threshold** / **Ratio** / **Attack** / **Release** / **Make-Up** / **Mix** / **SC HPF** — 7 knobs across the top centre.
- **Bus Comp In** — single button enabling the BC section.

The BC controls drive the BC Instance on the **BC anchor track** — the track UC1 has currently pinned for Bus-Comp display. Encoder 2 (the *Secondary encoder* right of the central screen) scrolls the anchor between BC-bearing tracks.

The mechanical BC VU meter is driven from REAPER via the PreSonus standard `GainReduction_dB` host-side hook. Rest position = bottom of scale; the needle swings up through GR magnitude.

8.5 Central control panel (between CS and BC)

A column of buttons + the central LCD + the two encoders:

- **Back** / **Confirm** — navigate the on-screen menus (Routing / Presets / etc.).
- **Routing** — opens the Routing menu on the LCD.
- **Presets** — opens the Presets menu.
- **360°** — default `mixer_toggle` (open / close Rea-Sixty Settings; bindable on its own UC1 entry so it can diverge from the UF8 360° key).
- **Magnifier** — no factory action; bindable.

8.6 CHANNEL encoder (left of the central LCD)

The large rotary on the central control panel.

- **Rotation** — Channel Select (move REAPER track selection ±). The surrounding 7-segment digit shows the absolute REAPER track number of the focused track.
- **Push** — push event arrives as button 0x0D; default binding empty.
- When **Settings → Modes → FX / Cycle → “UC1 Encoder 1 (CHANNEL)”** is ticked AND a cycle-kind Selection Mode is engaged, rotation drives `reasixty_dispatchSelModeCycle` instead.

8.7 Secondary encoder (right of the central LCD)

ButtonId `Uc1Encoder2` in the bindings system; SSL calls it the “Secondary” or “BC” encoder.

- **Rotation** — `bc_track_scroll` by default (scroll the BC anchor between BC-bearing tracks). Rebindable, including to `instance_cycle` or `fx_cycle`.

- **Push** — `show_focused_plugin_gui` by default (toggle floating window of the cursor instance from the most recent cycle). Rebindable.
- Cycle-Control mask includes “UC1 Encoder 2 (BC)” — same SEL-Mode override mechanism as above when ticked + a cycle-kind Selection Mode active.

8.8 Central LCD zones

Three addressable zones:

- **Channel-Strip readout (zone 0x03)** — last-touched CS parameter name + value, e.g. “HF Gain +3.0 dB”. Fades after a few seconds.
- **Bus-Comp readout (zone 0x05)** — last-touched BC parameter name + value.
- **Central main (zone 0x0F)** — multi-overlay area. Shows: track-name header + focused Instance variant (default), or the prev / curr / next Instance-carousel triple (after an Instance / FX Cycle just fired), or the BC compressor mode / status, or the Nav-Mode markers/regions carousel. Overlays are mutually exclusive; the most recent claim wins.

8.9 Brightness

Set independently per channel (UC1 LEDs / UC1 LCDs / UF8 LEDs / UF8 LCDs) under Settings → Device → Brightness. Six `brightness_*_up` / `brightness_*_down` builtins drive these from a hardware button (LEDs / LCDs / Both, up / down).

8.10 UC1 GR Calibration

If the UC1’s mechanical VU meter or the CS Dynamics GR LEDs drift from their printed scale, a per-tick offset table at the very bottom of **Settings → Device** corrects each printed dB tick individually. The workflow mirrors SSL 360°’s own BC VU calibration tool — click `Test` next to a tick, then + / – until the UC1 lines up with the printed marking. Auto-saved per-tick. `Stop test` resumes normal GR.

9 Settings window

The Settings window is a dockable RealmGui context. Open with the `360°` key (default), the `mixer_toggle` builtin, or REAPER's Action `Rea-Sixty: Toggle Settings window`.

Eight sidebar tabs: Device · Appearance · Bindings · Modes · FX Learn · Selection Sets · Parameter Groups · About.

9.1 Device pane

9.1.1 Connected devices

Live status dots for UF8 and UC1 (`[connected]` / `[not connected]`), plus the serial number when connected. Each device has an **Identify** button that overrides its LCD with a "THIS UNIT" marker for ~2 s — useful for multi-UF8 setups (deferred — see *Pending* at the bottom of the pane).

9.1.2 Brightness

Two 5-step sliders (Dark / Dim / Half / Bright / Full):

- **LEDs** — drives buttons + V-Pot rings + UC1 LEDs
- **LCDs** — drives the UF8 LCD strips + UC1 LCD + UC1 status displays

Set independently so you can crank the displays while keeping the LED ring dim, or vice versa. Six `brightness_*_up` / `brightness_*_down` builtins drive these from any button binding.

9.1.3 Display behaviour

Toggle	Effect
SEL LED follows REAPER track colour	Each strip's SEL LED renders the track's REAPER colour instead of monochrome. Off → SEL is white when selected.
GR meter source (combo)	*Only Show Channel Strip GR* — meter limited to SSL CS / mapped CS plug-ins. *Show any GR Data* — falls back to any FX exposing the PreSonus `GainReduction_dB` host-extension on the focused track (ReaComp, FabFilter, etc.).
Track selection follows parameter change	V-Pot / CS / BC knob edits on a non-selected track auto-select that track.
Touch selects channel	Touching a UF8 fader exclusively selects that strip's track.
SSL Strip Mode follows focused plugin window	When REAPER's last-focused FX is a CS Instance, SSL Strip Mode auto-engages.

Toggle	Effect
Plugin GUI follows active Instance	When an Instance Cycle / FX Cycle lands on a new target, an already-open floating plug-in GUI re-points to the new target.
Pin plug-in GUI position	Capture an XY coordinate (drag a window, click *Capture current*). Every subsequent managed `TrackFX_Show` snaps the window to that pin. Alternatively *Center on Screen*.
Pin FX-chain GUI position	Same pattern for FX-chain windows. Title matching looks for "FX:" on macOS.
Meter ballistic (combo)	Peak / VU / RMS — applies to the strip-bar level meters.

9.1.4 Tracks

Toggle	Effect
TCP follows UF8 selection	UF8-triggered track selection scrolls REAPER's arrange-view track panel (action 40913). MCP follow is always on.
Show tracks hidden in TCP	Off (default) → tracks REAPER has hidden in TCP also disappear from the UF8.
Show tracks hidden in MCP	Same, independently, for MCP hidden tracks.
Hide tracks in collapsed folders	Independent surface-side mirror of REAPER's "hide children of collapsed folders" preference. When on, any track whose ancestor folder has `I_FOLDERCOMPACT == 2` (fully collapsed) drops from the UF8 strip list. Walks ancestors on every track-list rebuild so the filter follows live folder-state changes. Default off.

Toggle	Effect
Long track-name handling (combo)	How track names longer than the 7-char scribble-strip slot are shortened. *Truncate* (default) keeps the legacy first-7-chars cut ("Background Vocals" → "Backgro"). *Smart abbreviate* drops separators, then vowels after the first letter of each token, then collapses repeated consonants, then proportionally distributes the remaining char budget across tokens ("Background Vocals" → "BckgVcl", "Drums Bus" → "DrmsBs"). Short all-caps tokens (DI / FX / EQ / …) survive untouched. Mode switch repaints all 8 strips immediately.

9.1.5 Plug-ins

Toggle	Effect
Don't show offline FX	Cycle rings (FX Cycle, Instance Cycle, per-strip variants) and the UF8 colour-bar default cursor skip TrackFX slots whose `TrackFX_GetOffline` returns true. Offline-only tracks show a –.
Wrap Plugin Cycle	Default on (legacy behaviour) — cycle rings wrap from last FX back to first. When off, both ends of the FX chain hard-stop on every cycle path (Channel-Encoder FX/Instance Cycle, per-strip V-Pot FX/Instance Cycle), and the UC1 carousel shows no neighbour name past the first/last FX.
Auto-engage UF8 Plugin Mode for UF8-mapped plug-ins	When SEL-Mode cycle V-Pot push OR a `show_focused_plugin_gui` binding lands on a UF8-mapped plug-in, also engage UF8 Plugin Mode with GUI.

9.1.6 Keyboard Options

Toggle	Effect
Alt/Option + fader drag → snap back to original on release	Hold Alt/Option while moving a fader; release while still holding Alt → fader snaps back to its touch-on value. Mirrors REAPER's mouse Alt-drag.
Keyboard Shift acts as Shift modifier	When on, holding **Shift** on the host keyboard counts as the Shift modifier for any binding's Plain/Shift/Cmd/Ctrl modifier slot — in addition to the hardware `mod_*` bindings.
Keyboard Cmd acts as Cmd modifier	Same, for Cmd on macOS.
Keyboard Ctrl acts as Ctrl modifier	Same, for Ctrl on Windows / Linux.
Shift activates Fine mode (V-Pots / encoders, not faders)	When on, holding the Shift modifier (keyboard Shift, UF8 `FINE` key, or UC1 `Fine` button) drops V-Pot + encoder step size to ×0.25 for momentary fine resolution. Faders are deliberately excluded (they already have Alt-drag for fine control). Stacks with the UC1 `Fine` toggle. Off by default.

9.1.7 Pending

A single line documents currently-deferred features (multi-UF8 drag-to-reorder).

9.1.8 UC1 GR calibration

Per-tick offset editor for the mechanical BC VU meter and the CS Dynamics GR LEDs. See chapter UC1 hardware → UC1 GR Calibration.

9.2 Appearance pane

9.2.1 Theme

Three-way radio:

- **Vanilla** (default) — ReaImGui's built-in dark theme, no overlay.
- **Dark** — Rea-Sixty's themed dark palette (blue-grey base + soft accents).
- **Light** — the Light counterpart for users on a light-mode REAPER setup.

Re-themes every Settings panel + the FX Learn schematic. Hardware-face colours (UF8 silk-screen mockups, UC1 schematic) stay constant.

9.2.2 Font Size

Three-way radio: **Small** / **Normal** / **Large**. Drives every Settings widget except the UF8 / UC1 mockup schematic labels — those stay locked at 12 px so the schematics don't reflow when the picker changes. Numeric inputs (GR-cal table, FX Learn binding column) scale with the font picker so layouts stay aligned across sizes.

9.3 Bindings pane

Top of the pane: a tab bar with **UF8** and **UC1**, each rendering its hardware as a vector schematic. Click any button, knob, encoder, or fader to select it; the per-button editor opens below the schematic.

The "current layer" follows whichever Layer button (1 / 2 / 3) is highlighted in the schematic — click a Layer button to switch the live layer; the green outline indicates which one is active. There is no separate layer-tab strip.

Three click-to-edit special cases:

- **Top-soft-keys** (the 8 buttons above the V-Pots) open the **user-Quick slot editor** for the live (Layer, Quick, Sub-Bank) coordinate instead of the regular per-button editor — top-soft-keys are slot pickers, not direct actions.
- **Sub-bank selectors** (V-POT + 1..5) open the **sub-bank cell editor** with a Per-Quick LED override so the user can distinguish (Layer, Quick) contexts visually.
- Everything else uses the regular per-button binding editor.

9.3.1 Per-binding editor

For a regular button, the editor exposes:

- **Action type** — Native / REAPER Action / Keyboard / MIDI Command / Noop.
- **Action name** — text picker (auto-complete) for Native + REAPER Action.
- **Modifier slot** — Plain / Shift / Cmd / Ctrl. Each modifier slot is bound separately, so one physical button can carry up to 4 different bindings per layer.
- **Behavior** — Momentary / Toggle / Hold.
- **Long-press action** — separate action that fires after the long-press threshold (~500 ms) instead of the short-press action on release.
- **LED appearance** — colour + brightness override that replaces the action's default state-of mapping.
- For **MIDI Command** bindings: channel / note number / velocity / CC value as appropriate.
- For **REAPER Action** bindings: a search dialog that browses REAPER's Action List by name or command ID (also exposes ReaScript loading).

9.3.2 Right-click context menu

Right-clicking a button in the schematic opens Copy / Paste / Clear options for that binding.

9.3.3 Export / Import / Reset

Bindings are bundled into the **Setup** export available from the **About** pane (single file covers bindings + plug-in maps + Settings preferences + Parameter Group slot names). Per-binding-file export does not exist as a Bindings-pane action.

Bindings storage paths:

- macOS: `~/Library/Application Support/REAPER/rea_sixty/bindings.json`
- Windows: `%APPDATA%\REAPER\rea_sixty\bindings.json`
- Linux: `~/.config/REAPER/rea_sixty/bindings.json`

9.4 Modes pane

This pane configures how the Selection Modes behave when they are active. Each sub-tab corresponds to one Selection Mode (or one cycle behaviour) — switching Selection Modes on the hardware uses the modes themselves; this pane only sets their per-mode options.

Four sub-tabs: AUTO · FX / Cycle · REC · NAV.

9.4.1 AUTO

Setting	Effect
Show only tracks armed for automation writing (hide Trim / Read)	While AUTO Selection Mode is engaged, tracks in mode 0 (Trim) or 1 (Read) are hidden from the surface. Touch / Write / Latch / Latch-Preview tracks remain visible.
Fill from left / Fill from right (radio pair)	When fewer visible tracks than the 8 hardware strips, choose which side they collect on. Project order is preserved either way. Active only while AUTO Selection Mode is engaged.
Selection-Set Auto-Mode (combo)	`None` / `Trim/Off` / `Read` / `Touch` / `Write` / `Latch` / `Latch Preview`. When set, recalling a Selection Set in AUTO mode forces its member tracks into this REAPER automation mode. Deactivating the set (or leaving AUTO mode) reverts those tracks to Trim/Read (mode 0).

9.4.2 FX / Cycle

> Active only while a cycle-kind Selection Mode (FX Cycle or Instance Cycle) is on the V-Pot row. Picks which physical controls drive the cycle; the active FX opens in the chosen view (V-Pot push only).

Controls (multi-select checkboxes):

- UF8 V-Pots (per-strip cycle) — default ON
- UF8 Channel Encoder
- UC1 Encoder 1 (CHANNEL)
- UC1 Encoder 2 (BC)

V-Pots cycle per-strip (each strip's own track). The three single encoders cycle the focused track and override their normal function while SEL Mode is engaged.

V-Pot push opens active FX as (radio pair): Floating window / FX chain.

9.4.3 REC

Setting	Effect
Enable RME / TotalReaper integration	Master switch. Requires the TotalReaper extension. While REC Selection Mode is active, the assignments below dispatch TotalReaper actions against the strip's track.
V-Pot rotation → Preamp gain ±1 dB	Steps preamp gain instead of pan.
V-Pot rotation + Shift → Change input channel	Re-routes the strip's track input on rotation.
V-Pot push (combo)	TotalReaper action assignment. Choices: `None`, `Toggle 48V phantom`, `Toggle pad`, `Toggle phase invert`, `Toggle AutoLevel`.
Cut button (combo)	Same action list.
Solo button (combo)	Same action list.

(Polarity is not a separately assignable REC button — only V-Pot push / Cut / Solo.)

9.4.4 NAV

The NAV sub-tab is divided into five sections.

Activation — read-only list of which physical button currently fires each of the three Nav-mode toggle builtins:

- `marker_overlay_toggle` (Markers + Regions)
- `marker_overlay_markers_only_toggle`
- `marker_overlay_regions_only_toggle`

Each line shows the bound layer + button + modifier + long-press flag, or "(unbound)". Edit via Settings → Bindings.

View defaults

- **Default view on Nav Mode entry** (radio): `Regions` / `Markers in current region` / `Markers (all)` / `Last used`. Applied by `marker_overlay_toggle` only. *Markers in current region* snaps to the region under the playhead; falls back to Regions if the playhead is in a gap.
- **Region-press behaviour** (radio): `Jump + Drill` / `Jump only` / `Drill only`. Jump = move transport to region start; Drill = enter the region's marker list. RegionsOnly view-lock always suppresses Drill.

UF8 strip display

- **Lower-row format** (radio): `Off (V-Pot value)` / `Index (R03 / M07)` / `Timecode (MM:SS)`. Off keeps the V-Pot value visible; Index / Timecode overlay marker metadata on the lower row.

- **Color-bar source** (radio): `REAPER marker colour` / `Force palette grey`. REAPER honours the colour override set on each marker / region; Force grey suppresses it.

UC1 Encoder 2

- **Take over UC1 Encoder 2 while Nav Mode is active** (checkbox). When off, Encoder 2 rotation stays bound to its normal action (`bc_track_scroll` by default) and the UC1 LCD does not switch to the marker carousel — only UF8 reflects Nav Mode.
- **Carousel scope** — currently single option *Mirror UF8 view*. Independent UC1 scopes (Always Regions / Always Markers / Always Markers-in-UF8-region) are not yet implemented.

Push actions — Plain push, Shift + push, and Long-press each pick any of the same 7 actions via dropdown:

Action	Effect
Jump + Drill	In Regions view: jump transport to the region start AND enter the region's marker list. In Markers view: jump only (drill is meaningless there).
Jump only	Move transport / edit cursor to the cursor item.
Drill only	Enter the region's marker list (Regions view only; no-op elsewhere).
Back	Return from a drilled-into Markers-in-Region view to Regions.
Toggle View	Flip between Regions and Markers (all).
Add marker at playhead	Insert an empty marker at the playhead (or edit cursor when stopped).
Disabled	No-op.

Defaults: Plain = Jump+Drill, Shift = Drill only, Long = Back.

View-locks (Markers-only / Regions-only) suppress **Drill only** specifically (Jump+Drill collapses to Jump only; Drill only becomes a no-op). Every other action fires regardless of lock. Long-press threshold ~500 ms.

Auto-Follow

- **Auto-Follow playhead / edit cursor** (checkbox). While Nav Mode is active, the cursor strip tracks whichever marker / region the playhead is on (or the edit cursor when stopped). In Markers-in-Region view, the overlay auto-rolls into the next region when the playhead crosses out.

9.5 FX Learn pane

The FX Learn pane teaches third-party plug-ins to behave as virtual Channel-Strip or Bus-Comp Instances. Built-in maps (SSL CS 2 / 4K B/E/G / BC 2 / 360 Link) always win — user maps can't shadow them.

The pane has two views:

9.5.1 Master view (default)

Toolbar:

- **+ New** — opens the "new map" picker (browses your installed-FX catalog by name, lets you choose primary mode + UF8-Mode flag).
- **Export...** — write the full user-plug-in catalog to a JSON file.
- **Import...** — read a catalog JSON back in.

Below the toolbar:

- Search field (case-insensitive match against FX name or derived developer string).
- Table of user maps. Per row: display short, FX-name match, developer, domain, variant count, mapped-param count, last-edit date, **Edit** button, **Delete** button.

9.5.2 Editor view (entered via Edit)

Top bar:

- **Domain** picker — `Channel Strip` / `Bus Comp` / `None (UF8-only)`. UF8-only maps the FX into the per-strip view without claiming a CS/BC slot.
- **UF8 Mode** checkbox (UF8-only domain) — drives Instance Cycle / Plug-in Mode dispatch.
- **Primary mode** picker (CS variant family) and other domain-specific options.
- **Mockup toggle** — visualises the UC1 layout via a UC1 mockup PNG instead of the strip-bar schematic. Persisted in ExtState `ReaSixty/fxLearnMockup`.
- **AutoLearn** button — runs the pattern-matching engine (hardcoded SSL seeds + user-map dictionary; three-pass: exact / substring / token) against either the live FX on the focused track or the catalog's stored param snapshot. Confidence-scored suggestions open in an *AutoLearn Preview* modal with a per-row checkbox + confidence %, plus All / None bulk helpers. UF8 V-Pot suggestions auto-group by category (EQ / Comp / Gate / Filter / I-O / Misc). Accept applies every checked mapping into the active map.
- Breadcrumb **`← All maps`** to leave the editor.

Editor body — depends on the domain:

- **CS / BC domain:** the UC1 / strip schematic. Click a control to learn it to a plug-in parameter; the active row's combo lists every plug-in parameter (with current value if a live instance is on the focused track).

- **None (UF8-only):** the UF8 strip-bar schematic. Drag a strip slot (V-Pot, top soft-key, etc.) onto a plug-in parameter.

9.5.3 Right-click context menu

Right-clicking a mapped control on the UF8 / UC1 schematic opens per-control options:

- **Copy / Paste / Clear** the binding.
- **Fill sequential (right)** on a V-Pot / Fader / Solo / Cut / Sel — propagates the source-strip's attributes onto every strip to the right. Carried fields: faderInverted; V-Pot inverted / vpotMode / polarity / defaultNorm / stripColour / travel (range + curve + sensitivity); Solo / Cut / Sel colour; Reverse LED flag.
- **Inverted [off/on]** on a Fader / V-Pot — flips the rotation / direction-to-value mapping.
- **Reverse LED [off/on]** on a Solo / Cut / Sel button — XORs the LED on/off bit before painting. Use this for plug-ins whose Cut/Bypass param reports `1 = inactive` so the LED would otherwise stay bright while the function is off. Saved per `(fader-bank, strip, button)` in `user_plugins.json`.
- **V-Pot mode: Value / Toggle** on a V-Pot — Value = continuous (rotate scrubs, push resets to *Push reset*); Toggle = binary (rotate ignored, push flips 0↔1).
- **Polarity: Unipolar / Bipolar** on a Value-mode V-Pot — Unipolar (default) renders the LCD ring as L→R sweep; Bipolar renders centre-out (like SSL Pan) and makes the Log / Exp curve presets mirror around 0.5. Made for Pan, EQ-gain, mid-range freq sweeps — anything where "neutral" sits in the middle.
- **Knob travel** (V-Pot only — see *Knob travel + curve editor* below) — inline Min / Max sliders + **Advanced…** opens the curve editor.
- **Push reset** slider (Value-mode V-Pot) — the value the V-Pot snaps to when pushed. On a Bipolar V-Pot, a small "0.5" quick-set button + hint appears when the slider isn't already at centre.
- **Display label** (inline text field) — per-slot override for the scribble-strip name (1..7 ASCII chars). Empty = falls back to the parameter's default short name. Persisted as `UserLinkSlot.customLabel` (FX-Learn slot) or `UserUf8BankSlot.label` (UF8 V-Pot) / `UserUf8StripBinding.faderLabel` (UF8 fader) in `user_plugins.json` — no schema bump.

9.5.4 Knob travel + curve editor

Every user-learned FX-Learn slot and every UF8 V-Pot binding can carry a custom range, response curve, and encoder sensitivity. Defaults (Min=0, Max=1, sensitivity=1, no curve) make the maths byte-identical to a plain linear mapping, so untouched bindings behave exactly as if the feature wasn't there.

In the per-slot right-click menu:

- **Min** / **Max** rows — a fixed-width 4-column table (label · slider · numeric input · **Set** button) so

both rows line up pixel-perfect. Slider scrubs in 0..1; the input accepts exact values; **Set** snaps the edge to the plug-in's *current* parameter value (handy when you've dialled the FX to where you want the limit and just want to capture it). Sliders auto-correct the opposing edge to keep Min ≤ Max.
- **Reset** — restores Min=0, Max=1.
- **Advanced...** — opens the Curve editor popup.

The Curve editor popup:

- **Sensitivity** — a single labelled row (label on its own line, then slider + numeric input + **1x** reset button). Range 0.1× .. 4×, encoder-delta multiplier. Combines multiplicatively with Shift = Fine (Shift still quarters on top of the user-set value). Hidden when editing a fader target.
- **Canvas** — draw a piecewise-linear response curve. Click empty space to add a breakpoint, drag to move, right-click to remove. The Y axis is normalised within [Min..Max], so the Linear preset is always a 45° diagonal regardless of how the range is trimmed.
- **Presets** — **Linear** (clears all breakpoints), **Log** (param rushes to the top — fine control near 0; on a Bipolar V-Pot the curve mirrors around 0.5 for a gentle ramp near centre + coarse at the edges), **Exp** (param stays small longer — fine control near 1; Bipolar mirrors for fine control at centre + rush to extremes), **Reset all** (clears curve + resets sensitivity to 1×). Bipolar polarity is re-read on every preset click so flipping it in the parent menu takes effect without re-opening the editor.
- **Close** dismisses the popup; all edits persist live as you make them.

9.5.5 Stepped parameters

When the bound parameter is a discrete-stepped enum (e.g. PSP Townhouse attack/release time selectors, HPF slope pickers, oversampling toggles) — anything REAPER reports with a non-zero per-step size via `TrackFX_GetParameterStepSizes` — the editor automatically switches to a stepped-aware layout. Sensitivity, range, and push-reset still apply; curve does not.

- **Sensitivity** label flips to **Detent speed**. The default 1.0× means *2 detents per step* on V-Pots and UC1 knobs (matches the legacy UC1 stepped feel). 2.0× → 1 detent per step. 0.5× → 4 detents per step. 4.0× and above fire multiple steps per detent. Shift-Fine still applies on top — at slider min (0.1×) Shift gives an effective 0.025× (≈40 detents per step) for ultra-precise crawling.
- **Canvas + preset row hidden.** A single info line replaces them: `Stepped parameter — N values (~X.XXX per step). Curve disabled; Min/Max snap to the step grid.`
- **Min / Max** in the right-click menu snap on commit to the nearest step boundary, so the encoder always traverses real values. Below the table a hint line reads `~K steps reachable in this range.`
- **Push reset** (Value-mode V-Pots) snaps the chosen default to the nearest step.

The runtime accumulator decays after 150 ms of inactivity, so reversing direction at sub-threshold doesn't leave residual fractional steps fighting the new motion.

In the FX-Learn schematic, slots with customised knob travel show:

- Two radial ticks at the Min / Max angles on the on-screen knob (7 o'clock → 12 → 5).
- A small centre dot when a curve is set.

UF8 V-Pots dispatch the math at the encoder-delta site (`sensitivity → inverseCurve → step → applyCurve`) so external automation writes stay coherent. UC1 channel-strip / bus-comp knobs and the EXT_FUNCS encoder honour the same path — a UC1 knob and a UF8 V-Pot bound to the same parameter stay in lock-step. Built-in SSL CS / BC slots are intentionally untouched and keep the legacy linear + EQ-gain virtual-notch path; knob travel only kicks in when a user-learned slot is present for the focused plug-in.

> **UF8 faders intentionally exclude knob travel.** Absolute-position + motor feedback creates round-trip races with plug-in quantisation (fader jumps during user motion, snaps on release). The plug-in's own taper is the right place for fader-side shaping.

9.5.6 Multi-instance picker

When the focused track has multiple FX matching the map's name, the editor surfaces a combo to choose which Instance's live readouts feed the editor. Picked index is per plug-in.

9.5.7 GR meter override

Default GR meter behaviour is to read the host-extension `GainReduction_dB` value REAPER exposes for any plug-in implementing the PreSonus VST3 convention. That works for most modern compressors out of the box, with no setup.

When a plug-in doesn't expose the host-extension (or exposes a wrong value), a small **GR** button next to **AutoLearn** in the editor header opens a compact override popup:

- Combo lists every VST3 parameter on the editing map; pick the one that reads the plug-in's gain-reduction value.
- **Offset (dB)** slider — added before |abs| at render time. Lets you calibrate compressors whose GR reads negative-going (e.g. -6 dB at peak reduction → set offset −6 so the meter reads +6).
- **(none)** / **Use host extension** clears the override and restores the default behaviour.

When set, the button shows a tick mark next to the **GR** label. The override flows through to both the UC1 BC VU motor calibration tables and the DYN GR LED strip. Per-map; saved in `user_plugins.json` under `metering.grVst3Param` + `metering.grOffsetDb`.

9.5.8 Param snapshot

When a catalog entry is created with the FX live on a track, parameter names + value formatters are snapshotted into the catalog so the editor stays usable even if no instance of the plug-in is currently loaded.

9.5.9 Storage

Catalog file at `~/Library/Application Support/REAPER/rea_sixty/user_plugins.json` (and equivalent paths on Windows / Linux). Versioned schema (currently v7). Old v5 / v6 files auto-migrate on first load.

9.6 Selection Sets pane

Eight slots (1..8). The slot acts as a filter — combined with any active Folder Mode / Show-Only-Selected / AUTO-mode filters (a track must pass all of them). Recall toggles — pressing the active slot's hardware button again deactivates it.

Each slot is either:

- **Snapshot** — fixed list of REAPER track GUIDs frozen at save time.
- **Group** — bound to a REAPER track group (1..64). Membership refreshes every onTimer tick from `GetSetTrackGroupMembership` across all Lead/Follow categories (ANY category).

The slot rows are laid out as a fixed-width 7-column table so columns align across rows regardless of slot type. Left to right:

- **`• Slot N`** — the • prefix marks the currently active slot.
- **Global** checkbox. When ON, the slot's content is workspace-global (ExtState, persists immediately). When OFF, project-scoped (ProjExtState, persists with the project save). Group slots benefit most from Global since "group N" is a stable concept across projects.
- **Type** combo: `Snapshot` / `Group`.
- **Name** text field.
- **Grp** spinner (Group rows) — REAPER track group index 1..64. Snapshot rows show `(N tracks)` in this column instead.
- **Save** button — Snapshot rows only. Overwrites the slot's GUID list with the current REAPER selection. (Save is hidden on Group rows — pressing it there would silently convert the slot to Snapshot and drop the live group binding, which is bad UX.)
- **Clear** button — Snapshot rows only. Empties the slot.

Recall is not a settings-pane button: slot activation lives on the hardware. Bind a button to the `Recall Selection Slot (toggle)` builtin with param 1..8 — pressing it engages the slot; pressing it again disengages. The • marker shows which slot is currently active.

9.6.1 Auto-mode binding

A single global "Selection-Set Auto-Mode" combo (Settings → Modes → AUTO) applies to every slot. When a slot is active in AUTO Selection Mode + the combo is set to a value other than `None`, recalling that slot arms its tracks to the selected automation mode. Leaving the slot (or switching out of AUTO) reverts them to Trim/Read.

9.6.2 Slot bank-snap

Recalling a slot snaps the surface to strip 0 = first channel of the set, so larger sets always start at the beginning. Re-pressing the same slot key keeps your current bank position.

9.6.3 Driving from hardware

Bind buttons to **Recall Selection Slot (toggle)** with param 1..8, and **Save current REAPER selection to slot** with param 1..8, via Settings → Bindings.

9.7 Parameter Groups pane

Parallel parameter control across multiple tracks: while a slot is active, plug-in tweaks on the focused track copy to every member track of the slot.

Top of the pane:

- **Multi-Select acts as temporary Parameter Group** (checkbox). When on AND no persistent slot is active AND multiple tracks are selected, those tracks become the live group.

Slot rows are laid out as a fixed-width 6-column table (so columns stay aligned across rows). Per row, left to right:

- **`• Slot N`** — • prefix marks an active slot.
- **Active** checkbox — toggle this slot's active state. Multiple slots can be active simultaneously (the fan-out is union-of-all-active-slot-members).
- **Name** text field.
- Member count display: `(N members)`.
- **Add Selected** button — add currently-selected REAPER tracks to this slot's membership.
- **Clear** button — empty the slot's membership.

Bottom of the pane:

- **Remove Selected Tracks from All Groups** button — pull every currently-selected track out of every slot.
- Hint text pointing at the Param Group native actions in Settings → Bindings.

9.7.1 Storage

Per-track slot membership lives in `P_EXT:rea_sixty:param_groups` as an 8-bit bitmask (one bit per slot). Slot metadata (names, active flag) lives in a project-scoped JSON sidecar (`param_groups.json`).

9.8 About pane

The pane stacks several sections from top to bottom.

9.8.1 Header

- Title + tagline (“Open-source SSL 360 replacement for UF8 / UC1”).
- Author byline: “Made by Frank Acklin @ Stoersender Studio, Switzerland”.
- Button **stoersender-studio.ch** — opens the studio website in the system browser.
- Commit-count blurb (“This project took N commits so far”).
- “You can buy me a beer:” + **paypal.me/FrankAcklin** button.

9.8.2 Versions

- **Version** — `git describe --tags --always --dirty` of the source tree at build time. On a tagged release: `v0.1.8`. Past a tag: `v0.1.8-N-g<sha>` (N commits past the tag). With uncommitted changes: trailing `-dirty`. Read this line first when triaging issues so it’s obvious which build is loaded.
- **Build** — date + time of the compiled extension.
- **REAPER** — the host REAPER version string.
- **ReaImGui** — the bundled-ABI banner (currently v0.10).

9.8.3 Project

- Repository URL display + **Open repository in browser** button.

9.8.4 Setup (Export / Import / Reset)

A single bundled file format covering: bindings, learned plug-in maps, Parameter Group slot names, and every Settings preference. *Selection Sets + Parameter Group track memberships stay per-project and travel with the .RPP.*

- **Export setup…** — save the bundle to a chosen JSON file.
- **Import setup…** — load a bundle (replaces in-memory state immediately; warnings reported inline).
- **Reset to factory defaults** — confirmation popup. Replaces bindings + learned FX + parameter-group slot meta + every Settings preference with the baked-in factory configuration. Per-project Selection Sets and Parameter Group memberships are untouched.

9.8.5 Windows USB driver (Windows only)

Section appears only when the build is Windows.

- Text: binds UF8 + UC1 to WinUSB so libusb can claim them without Zadig. One-time setup, requires admin. SSL 360° stops seeing the devices after install — reinstall SSL 360° to revert.

- **Install UF8/UC1 WinUSB driver** button — kicks off the in-product installer with a UAC + publisher prompt. After acceptance, unplug + replug devices.
- **Uninstall** button — runs `pnputil /delete-driver /uninstall` (UAC prompt), removes `rea_sixty_winusb.inf` from the driver store, and clears the Rea-Sixty signing cert from the My / Root / TrustedPublisher stores. After uninstall, unplug + replug devices; the SSL 360° driver (or whatever was previously bound) takes over again.

9.8.6 Linux udev rule (Linux only)

Section appears only when the build is Linux.

- Text: grants non-root USB access by installing `/etc/udev/rules.d/99-rea-sixty.rules`. One-time setup, requires sudo (graphical password prompt).
- **Install Linux udev rule** button — runs pkexec, writes the rule, reloads udev. After install, unplug + replug UF8 + UC1, then restart REAPER.
- **Uninstall** button — pkexec removes `/etc/udev/rules.d/99-rea-sixty.rules`, then reloads + triggers udev. After uninstall the surface drops back to root-only USB access until a rule is reinstalled.

9.8.7 Logs

- Lists the diagnostic log paths (`/tmp/reaper_uf8_frames.log`, `/tmp/reaper_uf8_colors.log`, etc.).
- **Reveal /tmp in Finder** button (macOS only — equivalent on other platforms TBD).

10 Native actions

Bind any of these to a UF8 / UC1 control in Settings → Bindings → *(button)* → Native. Actions that take a parameter (slot number, soft-key index, etc.) are flagged below.

10.1 Selection Mode toggles

Switch which Selection Mode is active (see chapter **Selection Modes** for what each one does). Pressing a mode's toggle while it is already active returns to NORM.

- **`selection_mode_norm`** — set the active Selection Mode to NORM (the default).
- **`selection_mode_rec`** — toggle REC mode (SEL = arm).
- **`selection_mode_rec_mon`** — toggle REC+MON mode (SEL = arm + monitor).
- **`selection_mode_auto`** — toggle AUTO mode (SEL = step the track's automation mode; V-Pots show automation indicator).
- **`selection_mode_instance`** — toggle FX Cycle Sel-Mode (V-Pots walk every FX per strip; push opens active FX).
- **`selection_mode_instance_cycle`** — toggle Instance Cycle Sel-Mode (V-Pots walk only Instances per strip).
- **`selection_clear_all`** — clear the REAPER track selection.

10.2 Channel Encoder mode toggles

Change which job the large CHANNEL encoder does. The current mode persists across REAPER restarts.

- **`encoder_nav`** — Channel Select (the default; rotation moves track selection ±1).
- **`encoder_nudge`** — playhead nudge (step size from REAPER's nudge setting).
- **`encoder_focus`** — synthesised mouse-wheel under the screen cursor (use to scroll plug-in windows, browsers, etc.).
- **`encoder_markers`** — step prev / next REAPER marker.
- **`encoder_bank_by_1`** — surface bank by 1 strip per detent.
- **`encoder_last_param`** — step the last-touched REAPER parameter ±.
- **`encoder_instance`** — Instance Cycle on the focused track.
- **`encoder_fx_cycle`** — FX Cycle (every FX) on the focused track.
- **`encoder_selset_cycle`** — step through populated Selection Set slots (off → 1 → 2 → … → off).
- **`encoder_mode_dispatch`** — rotation handler that routes to whichever encoder mode is currently set. Bound by default to the CHANNEL encoder so rotation just "does the right thing"; rebind if you want a fixed behaviour.

10.3 Direct encoder rotation handlers

Bind these to a non-CHANNEL rotation (UC1 Encoder 2, footswitches with rotation, etc.) when you want a single fixed behaviour regardless of the global encoder mode.

- **`instance_cycle`** — Instance Cycle on focused track (rotation = step ±).
- **`fx_cycle`** — FX Cycle on focused track.
- **`select_relative`** — step REAPER track selection ±1.
- **`playhead_nudge`** — playhead nudge ±.
- **`mouse_scroll`** — synthesised scroll-wheel under the screen cursor.
- **`bc_track_scroll`** — scroll the UC1's Bus Comp anchor track ±1 (which track the BC section is pinned to).

10.4 Plug-in Mixer modes

Engage / disengage the on-surface plug-in editing modes.

- **`ssl_strip_mode_toggle`** — toggle SSL Strip Mode. While active, fader → CS Fader Level, V-Pots → CS controls, soft-keys → CS bypass / EQ-In / Filter-In.
- **`ssl_strip_mode_toggle_with_gui`** — same, AND open the CS plug-in's floating GUI (pinned per the GUI-pin settings).
- **`uf8_plugin_mode_toggle`** — toggle UF8 Plug-in Mode. All 8 strips become the strips of a single FX-Learn-mapped plug-in. Bank ← / → flips between fader-banks A / B for ≥9-control plug-ins.
- **`uf8_plugin_mode_toggle_with_gui`** — same, AND open the plug-in's floating GUI.

10.5 Plug-in commands

These act on the FX the cursor currently points at on the focused track (the FX that the last cycle landed on; defaults to the first online Instance after a fresh load).

- **`show_focused_plugin_gui`** — toggle the floating GUI of the cursor FX. With the Device option *Auto-engage UF8 Plugin Mode* on, also engages UF8 Plug-in Mode if the cursor lands on a UF8-mapped plug-in.
- **`plugin_bypass`** — toggle bypass of the cursor FX.
- **`plugin_offline`** — toggle offline state of the cursor FX.
- **`plugin_preset_next`** — load the next preset.
- **`plugin_preset_prev`** — load the previous preset.
- **`plugin_preset_cycle`** — encoder-rotation variant that steps presets ± by detent count.
- **`plugin_move_up`** — move the cursor FX up one slot in the track's FX chain.
- **`plugin_move_down`** — move the cursor FX down one slot.
- **`show_fx_chain`** — open / close REAPER's FX chain window for the focused track (pinned per the FX-chain pin settings).
- **`close_all_fx_guis`** — close every floating FX window in the project.

- **`fx_param_inc`** — step the FX-Learn slot a V-Pot is bound to upward from a button. Action-picker exposes the slot target (combo built from the built-in PluginMap registry — link IDs are stable across SSL CS / BC variants), a step-size slider, and a wrap-vs-clamp checkbox. Honours the slot's range, curve, and sensitivity, so a button bound to `fx_param_inc` and a V-Pot bound to the same slot stay in sync. Useful for "+1 dB" or "next preset value" buttons.
- **`fx_param_dec`** — same as `fx_param_inc` with the sign flipped.

10.6 Instance navigation

These step *only* the Instance index (CS / BC / UF8-Mode-mapped). They are the focused-domain equivalent of the Instance Cycle Sel-Mode, but as standalone bind targets.

- **`instance_next`** — next Instance in the focused domain on the focused track. Wraps.
- **`instance_prev`** — previous Instance in the focused domain.
- **`domain_cs`** — set the focused domain to Channel Strip (so subsequent `instance_next` walks CS Instances, UC1 CS section refreshes).
- **`domain_bc`** — set the focused domain to Bus Comp.

The focused parameter slot is **preserved across an Instance Cycle** when the new instance offers the same LinkSlot (same domain, same parameter convention). Stops the focused-param surfaces (UC1 BC/CS encoder, V-Pot mirroring) from snapping back to slot 0 — typically the Bypass / FX In toggle — on every cycle step. Cross-domain cycles (CS → BC) and Domain::None (UF8-only user maps) still reset to slot 0 because the slot index isn't meaningful there.

10.7 Per-track automation modes

Set the focused track's automation mode. (`auto_off` and `auto_trim` are alternate names for the same REAPER mode 0; both kept for binding-file compatibility.)

- **`auto_off`**, **`auto_trim`** — mode 0 (Off / Trim).
- **`auto_read`** — mode 1.
- **`auto_touch`** — mode 2.
- **`auto_write`** — mode 3.
- **`auto_latch`** — mode 4.
- **`auto_latch_prv`** — mode 5 (Latch Preview).

10.8 Project-global automation override

Same six modes, but applied via REAPER's *global override* (overrides every track without changing each track's own mode).

- **`auto_off_global`**, **`auto_trim_global`**, **`auto_read_global`**, **`auto_touch_global`**, **`auto_write_global`**, **`auto_latch_global`**, **`auto_latch_prv_global`**.
- **`automation_zero_all`** — reset every track's automation mode to Trim/Read (mode 0). Useful to revert a write session.

10.9 Bank navigation

- **`bank_left`** — scroll the surface 8 strips left. In UF8 Plug-in Mode the same button flips between fader-banks A / B (for 16-strip plug-ins) instead.
- **`bank_right`** — scroll 8 strips right (same fader-bank flip in UF8 Plug-in Mode).
- **`bank_by_1_left`** / **`bank_by_1_right`** — scroll one strip at a time.
- **`home`** — clear all routing toggles (send / receive views) so V-Pots and faders return to track volume + pan.
- **`page_left`** / **`page_right`** — step the SSL Soft-Key PAGE bank (prev / next of the 6 CS banks + 2 BC banks).

10.10 DAW Layer keys

- **`layer_select_1`** / **`layer_select_2`** / **`layer_select_3`** — switch SSL DAW layer.

10.11 SSL Soft-keys

- **`softkey_bank_select` (param: 0..5)** — select Soft-Key bank N (CS banks 0..5; in Bus-Comp mode, 0..1).
- **`ssl_softkey` (param: 0..7)** — fire SSL Soft-Key cell N in the currently selected bank.
- **`ssl_bank_vpot` (param: 0..7)** — fire SSL V-Pot N (bank-current). The native bindings the Top Soft-Keys use to map to the active CS/BC bank's parameters.

10.12 Send / Receive

- **`send_this` (param: 0..7)** — toggle the Send view for slot N. With the view active, V-Pots drive that send's level instead of pan.
- **`recv_this` (param: 0..7)** — same for Receive views.

10.13 Selection Sets

- **`selset_recall` (param: 1..8)** — toggle slot N: activate if inactive, deactivate if already active. Activation filters the surface to the slot's tracks and snaps the bank to strip 0 = first slot track.
- **`selset_save` (param: 1..8)** — save the current REAPER track selection into slot N.
- **`selset_cycle`** — encoder-rotation handler that steps off → first populated slot → next → ... → off. Skips empty slots.

10.14 Surface filters / view toggles

- **`folder_mode`** — toggle Folder Mode (only top-level tracks visible; folder children appear on spill).
- **`show_only_selected`** — toggle Show Only Selected (only currently-selected REAPER tracks appear).

- **`mixer_toggle`** — open / close the Rea-Sixty Settings window. Default binding for the `360°` key.

10.15 Nav overlay (Markers + Regions)

- **`marker_overlay_toggle`** — toggle the Nav overlay (UF8 strips show markers / regions, soft-keys jump to them).
- **`marker_overlay_markers_only_toggle`** — Nav overlay restricted to markers only.
- **`marker_overlay_regions_only_toggle`** — Nav overlay restricted to regions only.

10.16 Track arming

- **`tracks_arm_all`** — toggle arm on every track in the project. State is "all armed" ↔ "all un-armed"; mixed state arms everything first, then needs a second press to unarm all.

10.17 Brightness

Each press steps one level (Dark → Dim → Half → Bright → Full). The "Both" variants step LEDs + LCDs in lockstep.

- **`brightness_leds_up`** / **`brightness_leds_down`** — LED ring + button-LED brightness.
- **`brightness_lcds_up`** / **`brightness_lcds_down`** — UF8 LCD + UC1 LCD brightness.
- **`brightness_both_up`** / **`brightness_both_down`** — combined.

10.18 Zoom

Each maps to a REAPER zoom action (defaults for the UF8 cursor pad).

- **`zoom_up`** — zoom in vertically (action 40112).
- **`zoom_down`** — zoom out vertically (40111).
- **`zoom_left`** — zoom out horizontally (1011).
- **`zoom_right`** — zoom in horizontally (1012).
- **`zoom_center`** — zoom to fit the whole project (40295).

10.19 Parameter Groups

- **`param_group_remove_all`** — remove the currently-selected tracks from every Parameter Group slot they appear in.
- **`multi_select_as_temp_group_toggle`** — toggle the "multi-select acts as a temporary parameter group" behaviour. When on, any multi-track selection becomes the active group for parameter-fan-out; when off, the active group is whichever persistent slot is selected.

10.20 Modifier keys

When held, these shift every other binding to its modifier slot. The three matching UF8 keys (Shift / Cmd / Ctrl, or FINE for Shift) are bound to them by default but you can rebind to any other button to relocate the modifier.

- **`mod_shift`** — Shift modifier. Double-click latches (press once more to unlatch). The SSL `FINE` key uses this builtin.
- **`mod_cmd`** — Cmd modifier (macOS) / Windows key (Win) / Super (Linux).
- **`mod_ctrl`** — Ctrl modifier.

10.21 Surface-state toggles

- **`flip`** — swap fader and V-Pot values for the active routing target (e.g. swap send level and send pan between V-Pot ring and motorised fader).
- **`pan_force`** — force V-Pots to Pan regardless of the active Selection Mode. Escape hatch from cycle / REC / AUTO when you need pan back quickly.

10.22 Internal (not user-bindable)

- **`__reaper_action__`** — internal carrier the REAPER Action picker writes into a binding's action field when you select a REAPER action by name. The picker UI is what you use; this action name is just storage.

11 Plug-in Mixer modes

11.1 SSL Strip Mode

Engage with the `Plugin` button (default), via the `ssl_strip_mode_toggle` / `_with_gui` builtins, or via "SSL Strip Mode follows focused plugin window" auto-engage on a CS plug-in.

While SSL Strip Mode is on:

- Fader → CS Fader Level (the CS plug-in's own fader parameter, not REAPER's track volume)
- V-Pots → CS controls (per SSL's standard 6 soft-key banks)
- Soft-keys → CS bypass / EQ-In / Filter-In etc.
- The colour-bar Type zone shows the CS variant name (CS2, 4K B, 4K E, 4K G) or the user rename
- UC1 follows the same CS Instance

The `with_gui` variant also opens the CS plug-in's floating GUI alongside, pinning it via the pin-position settings.

`Page ←` / `Page →` step through the 6 SSL-defined Channel Strip soft-key banks. `Plug-in` again (or any other Plug-in Mode toggle) exits.

11.2 UF8 Plug-in Mode

Engage with the `uf8_plugin_mode_toggle` (or `_with_gui`) builtin. Default binding: long-press the Plug-in button.

While UF8 Plug-in Mode is on:

- All 8 strips become the strips of a single FX-Learn-mapped plug-in — addressable even when fewer than 8 REAPER tracks are visible (strips past the bank-track count source their content from the focused FX instead of going blank).
- Two fader banks (16 strips total for plug-ins that need more than 8 controls) — Bank ← / Bank → flips between A and B.
- 8 soft-keys above each strip drive the FX Learn-mapped TopSoftKey slot per bank.
- The active group of soft-key cells is the bank's "active TopSoftKey ring" — 1 of 8 ringed brightly.
- Unassigned banks act as no-ops (LEDs dim).

The active plug-in is the focused track's first UF8-Mode-mapped instance, or the Instance the cursor is currently on.

Per-V-Pot customisation lives in Settings → FX Learn → (right-click the V-Pot on the schematic). Each `(fader-bank, soft-key bank, strip)` V-Pot carries its own *V-Pot mode* (Value / Toggle), *Polarity* (Unipolar / Bipolar — Bipolar renders the LCD ring centre-out, like SSL Pan), *Push reset* value, *display label*, *strip colour*, and *knob travel* (Min / Max range + response curve + sensitivity — see *Knob travel + curve editor* under FX Learn pane). Fill Sequential propagates every one of these fields to the strips to the right.

11.3 360° Link

The SSL 360° Link plug-in (a wrapper that mirrors third-party VSTs into the 360° surface) is recognised natively. When the focused track has a 360° Link instance pointing at a learned plug-in, SSL Strip Mode / UF8 Plug-in Mode dispatch through the linked plug-in transparently.

User-renamed 360° Link instances show the rename instead of the generic “Link” / “L-BC” abbreviation.

12 REC + RME (TotalReaper) integration

Requires the **TotalReaper** extension (separate ReaPack package) and an RME interface supporting TotalMix FX 2.1 Global OSC.

Master switch: Settings → Modes → REC → "Enable RME / TotalReaper integration".

When on AND REC / REC+MON selection mode is active:

- V-Pot rotation → preamp gain ±1 dB (configurable per `V-Pot rotation → Preamp gain ±1 dB` toggle)
- V-Pot rotation + Shift → input channel reassignment (configurable per `V-Pot rotation + Shift → Change input channel` toggle)
- V-Pot push → action of choice (`None` / 48V / Pad / Phase invert / AutoLevel)
- Cut button → action of choice (same enum)
- Solo button → action of choice
- Polarity (if mapped) → action of choice

Strip colour bar shows the input channel name ("Mic 1", "Line 3") instead of the CS variant label.

Hardware inputs only — MIDI / multichannel inputs leave the original colour-bar label intact.

The TotalReaper action names are looked up via `NamedCommandLookup`; if TotalReaper isn't installed the integration silently no-ops.

13 Selection Sets

13.1 Active set as a track filter

When a slot is active, the visible track list is filtered to the slot's tracks. Filter is ANDed with Folder Mode / Show Only Selected so they compose naturally.

13.2 Group slots

A Group slot tracks REAPER's track-group membership in real time. Add a track to group N in REAPER → it appears in the surface; remove it → it disappears, without a re-recall.

13.3 Auto-mode interaction

`Selection-Set auto-mode` (Settings → Modes → AUTO) is a single global value: `-1` (off) or REAPER's automation mode index (0..5). When a selset is active in AUTO sel mode + the drop-down is non-off, recalling the slot arms its tracks to that mode. Leaving the slot OR leaving AUTO sel mode reverts those tracks to Trim/Read.

13.4 Persistence

Snapshot slots store GUIDs in REAPER's ProjExtState (per project). Group slots store: group index, project-scoped or global. The `Selection-Set auto-mode` value is project-global (ExtState).

14 Parameter Groups

Eight slots + a per-slot member list of REAPER tracks. The Active group's members all receive the same parameter edit when you twist a UF8 V-Pot, a UC1 knob, or move a fader on any one of them.

Slot management UI in Settings → Parameter Groups. Each slot:

- Name
- Member tracks (Add Selection, Remove, Clear)

Active group switch via the per-slot Active radio.

Temp-group mode: when `Multi-Select acts as Temp Group` is on, the active group is derived live from the current REAPER multi-track selection.

Unmapped FX deferred: if a member track lacks the same FX as the source track, the edit silently drops on that track.

Storage: each track's member-of-which-slots state is a bitmask in P_EXT (`P_EXT:rea_sixty:param_groups`). Slot metadata is in a project-scoped JSON sidecar.

15 Bindings

The Bindings tab renders the UF8 + UC1 hardware as schematics. Every button / knob / fader is editable.

15.1 Per-binding fields

- **Action type:** Native / REAPER Action / MIDI Command
- **Action name** (Native + REAPER Action) / MIDI message (MIDI Command)
- **Modifier:** None / Shift / Cmd / Ctrl / combinations
- **Trigger:** Press / Hold / Long-press
- **LED override:** colour + brightness

15.2 Modifier system

The three modifier-key builtins (`mod_shift`, `mod_cmd`, `mod_ctrl`) shift every other button's binding to that modifier slot while held. Modifier keys themselves are bindable to any physical button.

Double-clicking `Shift` (= the SSL `FINE` key) latches it on. Press once more to unlatch.

Modifier slots are per-modifier-combination, so a button can have up to 8 separate bindings.

15.3 Long-press

A long-press binding fires after the long-press threshold elapses. The short-press binding fires on release if the threshold wasn't reached. A button can have both bindings.

15.4 Toggle / Hold semantics

- **Press**: fires once per press edge
- **Hold**: keeps firing while held (for repeats like bank-shift)
- **Long-press**: fires once when the long-press threshold elapses

15.5 LED override

Every bound button can override its LED colour and brightness, independent of the action's natural state.

15.6 REAPER Action picker

Pick any action from REAPER's Action List by name or command ID. Supports filtering and ReaScript loading (Load ReaScript button drops a `.lua` file into the catalog).

Named commands (_-prefixed) resolve via REAPER's `NamedCommandLookup` so ReaScripts stay bound across re-numberings.

15.7 MIDI Command bindings

Emit any MIDI message: Note On/Off, CC, Program Change, Pitch Bend, NRPN. Channel / Note / Value editable. The message goes to a virtual MIDI port that the extension opens — route it in REAPER's MIDI plumbing as needed.

16 Operational modes (track filters)

16.1 Folder Mode

Toggle: `folder_mode` builtin. When on, only top-level (depth-0) tracks are visible on the surface. Folder children appear only when “spilled” — long-press a folder parent’s SEL button to toggle that parent’s spill.

16.2 Show Only Selected

Toggle: `show_only_selected` builtin. When on, only currently-selected tracks appear on the surface. Live filter — changing REAPER selection updates the surface within one timer tick.

16.3 Selection-Set filter

ANDs with Folder Mode / Show Only Selected. When a selset is active, only its tracks appear (further filtered by Folder Mode if also on).

16.4 Hide-hidden filter

Settings → Device → Tracks → “Show tracks hidden in TCP / MCP” (default both OFF) — hides tracks REAPER has hidden in TCP or MCP. ANDs with all other filters.

16.5 Auto-hide Trim/Read

Settings → Modes → AUTO → “Hide Trim/Read tracks while in AUTO mode” — within AUTO selection mode only, hides tracks whose automation mode is Off/Trim (0) or Read (1).

17 Troubleshooting

17.1 Rea-Sixty does not appear in Preferences → Control/OSC/Web → Add

The dylib / dll / so didn't load. Check:

- **macOS Console** for codesign / dyld errors.
- **Windows Event Viewer → Application** for module-load errors.
- **Linux:** start REAPER from a terminal and watch stderr.

Make sure the runtime libraries are in the same directory as `reaper_rea-sixty.{dylib,dll,so}` — for Windows this means `%APPDATA%\REAPER\UserPlugins\`, where the delay-load resolves them at first call.

17.2 Surface does not respond / "SSL360Core owns the device"

SSL 360° is running and has claimed the UF8/UC1 vendor interface exclusively. Quit SSL 360° and restart REAPER.

17.3 Disconnect after sleep / wake or sustained idle (macOS)

Known issue: both UC1 and UF8 IN endpoints can fail within ~3 ms of each other on a sustained host-side USB stack condition. `libusb_reset_device` does not escape it. Physical replug (one or both devices) recovers. Diagnostic logs in `/tmp/rea_sixty_uc1_stale.log` + `/tmp/rea_sixty_uf8_stale.log`.

17.4 Track-colour wrong

- Confirm Settings → Device → Display behaviour → "SEL LED follows REAPER track colour" is on.
- The colour-bar quantiser is HSV polar — subtle RGB differences in REAPER's colour picker can land on neighbouring palette slots. Pick a more saturated colour if the nearest palette slot looks wrong.

17.5 GR meter reads wrong on a third-party compressor

- Enable Settings → Device → Display behaviour → "GR meter source: Show any GR Data".
- Confirm the compressor implements the PreSonus VST3 host-side `GainReduction_dB` configparm. Plug-ins that don't expose this will not drive the meter regardless of settings.

17.6 Linux port resets / `xhci_hcd disabled by hub (EMI?)`

Plug UF8 and UC1 into separate PC USB ports. Daisy-chaining UC1 through UF8's downstream port triggers `xhci_hcd` port cycling on Linux 6.17.

17.7 Settings window doesn't appear

ReaImGui isn't installed. Install it via ReaPack. Hardware control still works without it.

17.8 Diagnostics

- `/tmp/rea_sixty_uc1_stale.log` — UC1 device-handle diagnostics
- `/tmp/rea_sixty_uf8_stale.log` — UF8 device-handle diagnostics
- macOS Console / Windows Event Viewer / Linux journal for in-process errors

18 Known limitations

- **Multi-UF8 support deferred.** Single-device assumption in bank-shift / colour-sync / VU-meter paths. The Connected devices list shows multi-UF8 setups but drag-to-reorder is greyed out.
- **No SSL Plug-in Mixer side panel** (the on-screen Mixer view alongside the Settings tabs) — Phase 2.6 on the roadmap, not in this build.
- **macOS: Intel binaries not shipped.** Apple Silicon only.
- **Layer 3 LED hardware quirk** on certain UF8 units — confirmed not fixable from code; layer functionality still works.
- **Firmware update breakage** is on the project to chase. Protocol is self-decoded; SSL gives no compatibility guarantees.

19 Vendor relationship

Not affiliated with Solid State Logic. SSL ACP Support replied to the project author on 2026-05-18 confirming “no objections to the public, open source project” but declined to share protocol documentation.

Protocol stays self-decoded; documented in `docs/protocol-notes.md` and adjacent capture notes in the repo.

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